

Wrist Kinematics of a Planar 2-DOF Arm

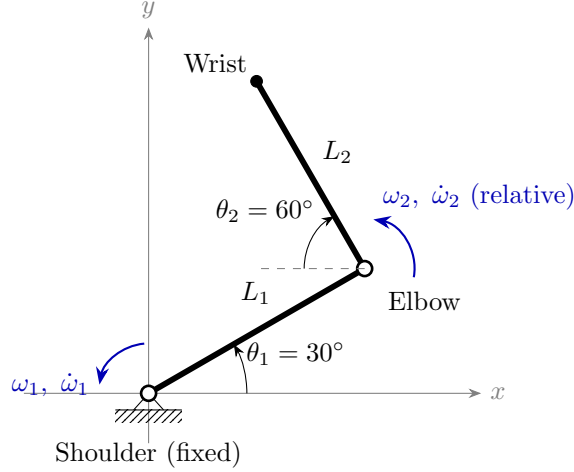


Figure 1: Initial configuration. θ_2 is measured from the $-x$ direction, so the forearm sits at 120° from $+x$.

i. Problem Statement

The problem is defined as follows: a two degree of freedom (2-DOF) robot arm with an upper arm length L_1 and a forearm length L_2 (meters) is fixed on a table via its shoulder joint. The shoulder pin is free to rotate, and has an angular velocity and acceleration of $\omega_1, \dot{\omega}_1$ respectively. The elbow joint rotates relative to the upper arm with an angular velocity ω_2 and an angular acceleration $\dot{\omega}_2$. The angular accelerations are both constant. At $t_0 = 0$, the upper arm is measured 30° from the $+x$ axis while the forearm is measured 60° from the $-x$ axis. Counterclockwise (CCW) rotation is taken as positive. The goal is to find the position, velocity, and acceleration of the wrist at a given point in time t .

ii. Coordinate Frames

First, the global coordinate frame, denote the O frame is positionally fixed on the arm shoulder with $\hat{i}, \hat{j}, \hat{k}$ unit vectors pointing in the x, y, z axis directions respectively, where the z axis points normal out of the page. In this paper, let o denote the origin of the global frame.

Another coordinate frame is placed on the rotating shoulder pin, denote the S frame, such that the unit vector $\hat{i}_s$ points along the upper arm link, $\hat{j}_s$ points 90° CCW from $\hat{i}_s$, and $\hat{k}_s$ points normal out of the page.

The third coordinate frame is placed on the rotating elbow joint, denote the E frame, such that the unit vector $\hat{i}_e$ points along the forearm link, $\hat{j}_e$ points

90° CCW from $\hat{i}_e$, and $\hat{k}_e$ points normal out of the page. Note that $\hat{k} = \hat{k}_s = \hat{k}_e$.

Observe that the shoulder and elbow frame unit vectors are rotations of the global unit vectors. Let $\theta_s = \theta_1$ and since θ_2 is measured clockwise from the $-x$ axis, we define $\theta_e = \pi - \theta_2$ for the initial state. So, we have

$$\begin{aligned}\hat{i}_s &= \begin{pmatrix} \cos \theta_s & -\sin \theta_s \\ \sin \theta_s & \cos \theta_s \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \cos \theta_s \hat{i} + \sin \theta_s \hat{j} \\ \hat{j}_s &= \begin{pmatrix} \cos \theta_s & -\sin \theta_s \\ \sin \theta_s & \cos \theta_s \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\sin \theta_s \hat{i} + \cos \theta_s \hat{j} \\ \hat{i}_e &= \begin{pmatrix} \cos \theta_e & -\sin \theta_e \\ \sin \theta_e & \cos \theta_e \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \cos \theta_e \hat{i} + \sin \theta_e \hat{j} \\ \hat{j}_e &= \begin{pmatrix} \cos \theta_e & -\sin \theta_e \\ \sin \theta_e & \cos \theta_e \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\sin \theta_e \hat{i} + \cos \theta_e \hat{j}\end{aligned}$$

Once an equation is developed for the velocity and acceleration of the wrist in terms of these unit vectors, we can apply these transformations to express them the O frame.

iii. Position

We first express the position of the wrist relative to the origin using the position of the elbow:

$$\vec{R}_{W/o} = \vec{R}_{E/o} + \vec{R}_{W/E}$$

Similarly, the relative position of the elbow to the shoulder is

$$\vec{R}_{E/o} = \vec{R}_{S/o} + \vec{R}_{E/S}$$

Since the shoulder joint is fixed to the O frame, we have $\vec{R}_{S/O} = \vec{0}$ and therefore by substitution:

$$\vec{R}_{W/o} = \vec{R}_{E/S} + \vec{R}_{W/E}$$

Using the coordinate unit vectors we defined in section (ii) we can write

$$\begin{aligned}\vec{R}_{E/O} &= \vec{R}_{E/S} = L_1 \hat{i}_s, \quad \vec{R}_{W/E} = L_2 \hat{i}_e \\ \implies \vec{R}_{W/o} &= L_1 \hat{i}_s + L_2 \hat{i}_e\end{aligned}$$

iv. Velocity

By differentiating the position expression with respect to time, we have

$$\vec{v}_{W/o} = \vec{v}_{E/o} + (\vec{v}_{W/E})_{rel} + \vec{\omega}_w \times \vec{R}_{W/E}$$

And likewise for the velocity of the elbow:

$$\vec{v}_{E/o} = \vec{v}_{S/o} + (\vec{v}_{E/S})_{rel} + \vec{\omega}_e \times \vec{R}_{E/S}$$

Because the arm links are rigid and each relative position vector has a constant length and a fixed direction within its own rotating frame and because the shoulder is fixed to the origin, we have

$$\vec{v}_{S/o} = 0, (\vec{v}_{E/S})_{rel} = 0, (\vec{v}_{W/E})_{rel} = 0$$

By substitution:

$$\begin{aligned}\vec{v}_{E/o} &= \vec{\omega}_e \times \vec{R}_{E/S} \\ \implies \vec{v}_{W/o} &= \vec{\omega}_e \times \vec{R}_{E/S} + \vec{\omega}_w \times \vec{R}_{W/E}\end{aligned}$$

Where $\vec{\omega}_e = \omega_1 \hat{k}_s$. Note that since ω_2 is *relative* to the shoulder pin's rotation ω_1 , then $\vec{\omega}_w = (\omega_1 + \omega_2) \hat{k}_e$. By substitution:

$$\begin{aligned}\vec{v}_{W/o} &= \omega_1 \hat{k}_s \times L_1 \hat{i}_s + (\omega_1 + \omega_2) \hat{k}_e \times L_2 \hat{i}_e \\ &= L_1 \omega_1 \hat{j}_s + L_2 (\omega_1 + \omega_2) \hat{j}_e\end{aligned}$$

So, our expression for wrist velocity before transforming the unit vectors to the global frame is $\vec{v}_{W/o} = L_1 \omega_1 \hat{j}_s + L_2 (\omega_1 + \omega_2) \hat{j}_e$.

v. Acceleration

By differentiating the full expression for velocity, we have our expanded expression for the wrist acceleration:

$$\vec{a}_{W/o} = \vec{a}_{E/o} + (\vec{a}_{W/E})_{rel} + 2\vec{\omega}_w \times (\vec{v}_{W/E})_{rel} + \vec{\dot{\omega}}_w \times \vec{R}_{W/E} + \vec{\omega}_w \times (\vec{\omega}_w \times \vec{R}_{W/E})$$

where

$$\vec{a}_{E/o} = \vec{a}_{S/o} + (\vec{a}_{E/S})_{rel} + 2\vec{\omega}_e \times (\vec{v}_{E/S})_{rel} + \vec{\dot{\omega}}_e \times \vec{R}_{E/S} + \vec{\omega}_e \times (\vec{\omega}_e \times \vec{R}_{E/S})$$

Similar to the reasoning discussed in section (iv), we have that

$$\vec{a}_{S/o} = (\vec{a}_{E/S})_{rel} = (\vec{a}_{W/E})_{rel} = 0$$

Recalling the fact that the relative velocity terms still are zero, we have that

$$\begin{aligned}\vec{a}_{E/o} &= \vec{\dot{\omega}}_e \times \vec{R}_{E/S} + \vec{\omega}_e \times (\vec{\omega}_e \times \vec{R}_{E/S}) \\ \vec{a}_{W/o} &= \vec{\dot{\omega}}_e \times \vec{R}_{E/S} + \vec{\omega}_e \times (\vec{\omega}_e \times \vec{R}_{E/S}) + \vec{\dot{\omega}}_w \times \vec{R}_{W/E} + \vec{\omega}_w \times (\vec{\omega}_w \times \vec{R}_{W/E})\end{aligned}$$

Now, we define $\vec{\dot{\omega}}_e = \dot{\omega}_1 \hat{k}_s$. However, for $\vec{\dot{\omega}}_w$ we need to observe that

$$\vec{\dot{\omega}}_w = \frac{d\vec{\omega}_w}{dt} = \vec{\dot{\omega}}_1 + \vec{\dot{\omega}}_2 + \vec{\omega}_1 \times \vec{\omega}_2$$

Since $\hat{k}_s = \hat{k}_e = \hat{k}$, we get that the cross product term evaluates to zero. Hence, $\vec{\dot{\omega}}_w = \vec{\dot{\omega}}_1 + \vec{\dot{\omega}}_2$. By substitution in the expression for wrist acceleration above,

we have by term:

$$\begin{aligned}
\vec{\omega}_e \times \vec{R}_{E/S} &= \dot{\omega}_1 \hat{k}_s \times L_1 \hat{i}_s = L_1 \dot{\omega}_1 \hat{j}_s \\
\vec{\omega}_e \times (\vec{\omega}_e \times \vec{R}_{E/S}) &= \omega_1 \hat{k}_s \times L_1 \omega_1 \hat{j}_s = -L_1 \omega_1^2 \hat{i}_s \\
\vec{\omega}_w \times \vec{R}_{W/E} &= (\dot{\omega}_1 + \dot{\omega}_2) \hat{k}_e \times L_2 \hat{i}_e = L_2 (\dot{\omega}_1 + \dot{\omega}_2) \hat{j}_e \\
\vec{\omega}_w \times (\vec{\omega}_w \times \vec{R}_{W/E}) &= (\omega_1 + \omega_2) \hat{k}_e \times L_2 (\omega_1 + \omega_2) \hat{j}_e = -L_2 (\omega_1 + \omega_2)^2 \hat{i}_e
\end{aligned}$$

Substituting these terms into the expression for $\vec{a}_{W/o}$ we get

$$\vec{a}_{W/o} = L_1 \dot{\omega}_1 \hat{j}_s - L_1 \omega_1^2 \hat{i}_s + L_2 (\dot{\omega}_1 + \dot{\omega}_2) \hat{j}_e - L_2 (\omega_1 + \omega_2)^2 \hat{i}_e$$

vi. Time Dependence

Now that we have an expressions for wrist position, velocity, and acceleration, we want to find how these quantities change with time. First, observe that all three are dependent on the angles θ_s, θ_e due to the unit vector transforms discussed in section (ii).

First, note that $\dot{\omega}_1, \dot{\omega}_2$ are constant. Now, denote the ω_1, ω_2 provided by the problem statement as $\omega_{1,0}, \omega_{2,0}$ to represent that they are initial quantities of the system. Then we have

$$\omega_1(t) = \omega_{1,0} + \dot{\omega}_1 t \quad \omega_2(t) = \omega_{2,0} + \dot{\omega}_2 t$$

Now, importantly θ_s, θ_e are changing with time, meaning the coordinate transform matrices are changing with time. The code evaluates these at every time step. First, denote $\theta_{s,0} = 30^\circ, \theta_{e,0} = 120^\circ$. Then, observe that θ_e changes with the absolute angular velocity and acceleration of the forearm. So, we have

$$\begin{aligned}
\theta_s(t) &= \theta_{s,0} + \omega_{1,0} t + \frac{1}{2} \dot{\omega}_1 t^2 \\
\theta_e(t) &= \theta_{e,0} + (\omega_{1,0} + \omega_{2,0}) t + \frac{1}{2} (\dot{\omega}_1 + \dot{\omega}_2) t^2
\end{aligned}$$

Substituting $\theta_s(t), \theta_e(t)$ into the unit vector transformations and $\omega_1(t), \omega_2(t)$ into the expressions of velocity and acceleration discussed in sections (iv,v) gives the wrist's position, velocity, and acceleration at any time t .

These expressions were implemented via Python (see repository) and used to create a program that visualizes the arm motion with different input variables. The Python script applies the coordinate transform at each discrete time step with the θ_s, θ_e calculated at that particular time step.

vii. Verification

Three tests were run on the Python-implemented solver to ensure the accuracy of the wrist position, velocity, and acceleration expressions, rewritten here prior

to unit vector transformations applied:

$$\begin{aligned}\vec{R}_{W/o} &= L_1 \hat{i}_s + L_2 \hat{i}_e \\ \vec{V}_{W/o} &= L_1 \omega_1 \hat{j}_s + L_2 (\omega_1 + \omega_2) \hat{j}_e \\ \vec{a}_{W/o} &= L_1 \dot{\omega}_1 \hat{j}_s - L_1 \omega_1^2 \hat{i}_s + L_2 (\dot{\omega}_1 + \dot{\omega}_2) \hat{j}_e - L_2 (\omega_1 + \omega_2)^2 \hat{i}_e\end{aligned}$$

The derived expressions are checked in three ways: against two limiting cases with known closed-form answers, and against a numerical approximation of the derivatives. All three compare the Python implementation to results that do not depend on the derivation above. These checks verify that the equations are solved correctly for the stated model; they do not validate the model against a physical arm, where physical properties and body forces apply.

Rigid rotation. Setting $\omega_{2,0} = \dot{\omega}_1 = \dot{\omega}_2 = 0$ locks the elbow, so the whole arm rotates as one rigid body about the shoulder at constant ω_1 . The wrist then moves in uniform circular motion with radius $r = |\vec{R}_{W/O}|$, so

$$|\vec{V}_{W/O}| = \omega_1 r \quad |\vec{a}_{W/O}| = \omega_1^2 r$$

The error was compared to a tolerance of $1 \cdot 10^{-10}$. For $L_1 = 0.35$ m, $L_2 = 0.30$ m, and $\omega_1 = 1.5$ rad/s, the wrist has $r = 0.4610$ m, $|\vec{v}| = 0.691$ m/s, and $|\vec{a}| = 1.037$ m/s². Over the full run, the maximum absolute deviations from these values were $3.886 \cdot 10^{-16}$, for position, $2.220 \cdot 10^{-16}$ m/s for velocity, and $2.220 \cdot 10^{-16}$ m/s² for acceleration, all three well within the set tolerance.

Shoulder held still. Setting $\omega_{1,0} = \dot{\omega}_1 = 0$ fixes the upper arm, so the elbow does not move and the wrist rotates about it. L_1, L_2 are the same as test 1. With $\omega_2(t) = \omega_{2,0} + \dot{\omega}_2 t$, the wrist undergoes circular motion of radius L_2 with changing speed, giving

$$|\vec{v}_{W/o}| = L_2 |\omega_2(t)|, \quad |\vec{a}_{W/o}| = L_2 \sqrt{\dot{\omega}_2^2 + \omega_2(t)^4},$$

where the acceleration combines the tangential component $L_2 \dot{\omega}_2$ and the centripetal component $L_2 \omega_2^2$. A nonzero $\dot{\omega}_2$ was used so that the time-dependent terms are exercised. The error was compared to a tolerance of $1 \cdot 10^{-10}$.

For $\omega_{2,0} = 1.5$ rad/s and $\dot{\omega}_2 = 0.3$ rad/s², the maximum absolute error were $1.110 \cdot 10^{-16}$ m/s and $4.441 \cdot 10^{-16}$ m/s², and the elbow position remained constant where max error was calculated 0 m.

Finite-difference check. The limiting cases test the equations with most inputs set to zero. To test them with all inputs active, the analytic results are compared with numerical derivatives. The numerical velocity $\vec{v}_N$ is obtained by differentiating the computed wrist position with np.gradient, and the numerical acceleration $\vec{a}_N$ by differentiating the analytic velocity similarly. The error was compared to a tolerance of $1 \cdot 10^{-4}$.

With all six inputs nonzero ($L_1 = 0.35, L_2 = 0.30, \omega_1 = 2, \omega_2 = -3, \dot{\omega}_1 = 0.2, \dot{\omega}_2 = 0.45$) and $\Delta t = 0.001$ s, the maximum absolute errors were $1.544 \cdot 10^{-6}$ m/s and $4.239 \cdot 10^{-6}$ m/s². When the test was ran with $\Delta t = 0.0005$, the errors were $3.861 \cdot 10^{-7}$ m/s and $1.060 \cdot 10^{-6}$ m/s². Hence, halving the time step reduced both errors by approximately a factor of 4.

Table 1: Verification results. Tests report maximum absolute error over the run; tests 1 and 2 use a tolerance of $1 \cdot 10^{-10}$ while test 3 uses a tolerance of $1 \cdot 10^{-4}$

Test	Check	Max error	Result
<i>1. Rigid rotation</i> ($L_1 = 0.30, L_2 = 0.35$ m, $\omega_1 = 1.5$ rad/s)			
	$ \vec{R}_{W/O} = \text{const.}$ ($r = 0.461$ m)	3.9×10^{-16} m	Pass
	$ \vec{V} = \omega_1 r = 0.691$ m/s	2.2×10^{-16} m/s	Pass
	$ \vec{a} = \omega_1^2 r = 1.037$ m/s ²	2.2×10^{-16} m/s ²	Pass
<i>2. Shoulder held still</i> ($\omega_{2,0} = 1.5$ rad/s, $\dot{\omega}_2 = 0.3$ rad/s ²)			
	Elbow stationary	0 m	Pass
	$ \vec{V} = L_2 \omega_2(t) $	1.1×10^{-16} m/s	Pass
	$ \vec{a} = L_2 \sqrt{\dot{\omega}_2^2 + \omega_2(t)^4}$	4.4×10^{-16} m/s ²	Pass
<i>3. Finite difference</i> (all inputs nonzero, $\Delta t = 0.001$ s)			
	Velocity	1.5×10^{-6} m/s	Pass
	Acceleration	4.2×10^{-6} m/s ²	Pass